A field-measurable regularized-Coulomb sliding law: the s_N(N) master curve, an N_c inversion, and a tidal-admittance ungrounding early-warning

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Abstract

Ice-stream sliding weakens as the bed approaches flotation, a behaviour large-scale models impose by hand (Joughin, Smith & Schoof 2019 add an ad-hoc near-flotation ramp to a regularized-Coulomb law). We show this weakening is a **closed-form pole** of the regularized-Coulomb law itself and, crucially, that it is **measurable from surface velocity alone**. Solving $\tau_b = CN(u/(u+u_0))^{1/m}$ at the driving stress gives, exactly, the sliding-sensitivity master curve $|s_N|(N) = m/(1-(N_c/N)^m)$ with $N_c = \tau_d/C$ — a simple pole at the flotation threshold N_c, verified to 1.4×10^{-4} against the numeric s_N. The curve makes three independent surface observables read the **same** law: a lake-drainage step ($du/u = |s_N| \cdot dN/N$), a continuous ocean-thermal-forcing gating slope, and the **tidal admittance** of velocity (whose 2f/1f harmonic ratio diverges toward N_c). A drainage-step population inverts for N_c to **0.2/0.4/1.0 %** at 5/10/20 % amplitude noise; the tidal probe recovers m and the dimensionless flotation proximity $R = (N_c/N)^m$ from velocity alone (answering the "no knowledge of N" objection), with the tidal forcing amplitude *measured*, not assumed (CATS2008 across the BedMachine grounding zone: median tidal pressure swing 0.30 % of overburden, rising to 0.45 % within 2–4 km of the line). Because the basal-drag stiffness $\partial\tau_b/\partial u \propto (1-R)^2/R \to 0$ at N_c, an ice stream nearing ungrounding should show rising variance and lag-1 autocorrelation of surface speed; we **operationalize** the established critical-slowing-down early-warning (Scheffer et al. 2009; Dakos et al. 2008; Rosier et al. 2021) as the continuous reading $R(t) = (N_c/N)^m \to 1$ from tidal admittance. Read through the same lens, three real MacAyeal-lake drainages sit far from the N_c pole and show no critical-slowing-down signature — a correct **true negative** on stable ice (the detectors fire on planted surges/CSD). The contribution is the field-measurable law and its inversion, not a new early-warning concept; absolute s_N remains uncalibrated and the strong test awaits a stream actually nearing flotation.

1. Introduction

The basal sliding law sets how fast an ice stream flows for a given driving stress and effective pressure $N = p_i - p_w$ (Schoof 2005; Gagliardini et al. 2007). Near the grounding line $N \to 0$ and sliding becomes Coulomb-limited; regularized-Coulomb laws (Joughin, Smith & Schoof 2019;

Tulaczyk et al. 2000; Gudmundsson 2011) capture this but introduce the near-flotation weakening through an *ad hoc* ramp and a fixed velocity scale u_0. The difficulty is that N is rarely known: it depends on the unobserved subglacial water pressure.

The tools we use, not claim, are the regularized-Coulomb friction law (Schoof 2005; Joughin et al. 2019), the nonlinear tidal (MSf) velocity response of ice streams (Gudmundsson 2011), and critical-slowing-down early-warning theory, *including its ice-sheet application* (Scheffer et al. 2009; Dakos et al. 2008; Rosier et al. 2021; Boers & Rypdal 2021).

What is new, stated against the closest prior work. Joughin, Smith & Schoof (2019) *impose* near-flotation weakening with a hand-tuned ramp; we *derive* it as a closed-form pole $|s_N| = m/(1-(N_c/N)^m)$ of the regularized-Coulomb law and turn N_c into a **measured** quantity. Rosier et al. (2021) established critical-slowing-down indicators (lag-1 autocorrelation, variance, Kendall-τ) for grounding-line flux ahead of marine-ice-sheet tipping; we do not re-claim the early-warning but give a specific **operational route** — reading $R(t) = (N_c/N)^m \to 1$ from the continuous tidal admittance of *surface velocity*, which needs no knowledge of N.

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2. The $s_N(N)$ master curve

2.1 $s_N(N)$ as the common observable. A lake drainage is an in-situ effective- pressure *step* whose surge amplitude $du/u = |s_N| \cdot (dN/N)$ measures the sliding sensitivity $s_N = dlnu_b/dlnN$; ocean thermal forcing lowers N *continuously*, so the gating slope $dlnu_*/dTF$ measures the **same** s_N ($+0.46 \to +0.62/^\circ C$, direct BedMachine N); ocean tides oscillate N at known amplitude, a third probe (§4). All three read one curve.

2.2 The closed-form master curve. Solving the regularized-Coulomb law $\tau_b = CN(u/(u+u_0))^{1/m}$ at $\tau_b = \tau_d$ gives, exactly, $|s_N|(N) = m/(1-(N_c/N)^m)$, $N_c = \tau_d/C$ — $\to m$ far from flotation, a **simple pole** at N_c ($\approx N_c/(N-N_c)$). Verified to 1.4×10^{-4} against the numeric s_N. This *derives* the near-flotation weakening that Joughin, Smith & Schoof (2019) impose by hand (ad hoc $h_{af} < h_T$ ramp + fixed u_0). (`validation/synthetic/sn_master_curve.py`)

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3. Inversion — measuring N_c and m

A drainage-step population recovers the flotation threshold N_c to **0.2 / 0.4 / 1.0 %** at 5 / 10 / 20 % amplitude noise; m is degenerate with the per-event drop far from flotation. The recovery is unbiased and sign-faithful under noise and null under permutation (synthetic plant-and-recover).

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4. The tidal third probe — N_c, m, and R from velocity alone

Ocean **tides** give a continuous probe: the fundamental velocity admittance equals $|s_N|$ and the 2f/1f harmonic ratio diverges toward N_c, so admittance + harmonics + the known tidal amplitude recover m and the dimensionless flotation proximity $R = (N_c/N)^m$ **from surface velocity alone** — a sliding-law reading of the Gudmundsson nonlinear MSf response, answering the “no knowledge of N” objection. The tidal forcing amplitude is **measured**, not assumed: sampling the CATS2008 tide model (USAP-DC 601235) across the BedMachine grounding zone (within 10 km of the line; $n = 1108$ valid cells) gives a tidal ocean-pressure swing with median **0.30 % of ice overburden** (median tide $\eta \approx 0.84m$), rising monotonically from 0.21 % at 8–10 km to 0.45 % at 2–4 km from the grounding line (`validation/synthetic/tidal_admittance_probe.py`, `validation/reports/tidal_forcing_gz.json`). **Contamination robustness (synthetic).** The two known non-sliding contributors at tidal frequencies — flexure leakage and tidally induced grounding-line migration, the mechanism Rosier & Gudmundsson (2020) identify as the only one strong enough to carry the observed MSf response, with Minchew et al. (2017) placing the strongest signal at the grounding zone — are quantified and screened rather than assumed away: a memoryless sliding response to a cosine tide is a Chebyshev expansion, so *every* velocity harmonic is phase-locked to the tide ($\phi_k \in \{0, \pi\}$; clean synthetic closure residual 10^{-15} rad). Lagged GL-migration rectification injects a 2f component that can fake flotation proximity (+46% bias on R at 2% modulation) but breaks 2f phase closure by 0.2–0.9 rad — flagged and screened before inversion; lagged (viscoelastic) flexure is flagged the same way at the fundamental, while exactly in-phase flexure is phase-invisible (20% R bias at 2% leakage) and is capped by station placement upstream of the flexure zone. Red (AR-1) noise from concurrent forcing at 1% of $\ln u$ over a 40-cycle record leaves ± 0.03 (68%) scatter on $R \approx 0.22$. (`validation/synthetic/tidal_admittance_contamination.py`.)

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5. An ungrounding early-warning (operationalizing established CSD theory)

The N_c fold makes the basal drag stiffness $\partial\tau_b/\partial u \propto (1-R)^2/R \to 0$ (critical slowing down), so an ice stream approaching ungrounding should show **rising variance and lag-1 autocorrelation of its surface speed** (Kendall-τ≈0.54 in an OU demonstration with N declining toward N_c). Critical-slowing-down early-warning is established (Scheffer et al. 2009; Dakos et al. 2008) and has already been applied to marine-ice-sheet tipping — Rosier et al. (2021) detected rising lag-1 autocorrelation and variance in grounding-line flux ahead of Pine Island tipping, and Boers & Rypdal (2021) in Greenland surface melt. The contribution here is **not** the early-warning concept but a specific **operational route**: reading $R(t) = (N_c/N)^m \to 1$ from the continuous tidal admittance of surface velocity. A single-snapshot **spatial** analogue (variance + along-flow correlation length rising toward the grounding line) is also derived (exact Lyapunov solve). **The discriminating prediction** is stated for the next near-flotation event: a stream approaching ungrounding must show the 2f/1f admittance ratio rising along the pole law $(\varepsilon/4)\, mR/(1-R) + ... \to \infty$ *together* with rising variance and lag-1 autocorrelation, with the phase-closure screen (§above) clean; observing the CSD indicators *without* the admittance divergence (or vice versa) falsifies the sliding-law reading. The natural watch list is the Amundsen sector (Thwaites and Pine Island grounding zones, where Rosier et al. 2021 place the tipping indicators and BedMachine puts the bed closest to flotation). (`validation/synthetic/spatial_ews.py`)

The fold exponents and a calibration-free spectral gauge (derived). The warning here is not generic: the s_N pole *is* the Schoof (2007) grounding-line saddle-node, so the critical-slowing-down exponents are fixed, not fitted. Near $N = N_c$ the restoring eigenvalue $\lambda \propto (1-R)^2/R \sim (N - N_c)^2$, giving $|s_N| \propto (N - N_c)^{-1}$, variance $\propto (N - N_c)^{-2}$ and $AC1 \to 1$ (recovered exponents $-1.00,\ -2.00,\ +2.00$). The same rate has a frequency-domain face: the velocity spectrum is a Lorentzian $S(\omega) = 2D/(\lambda^2 + \omega^2)$ whose corner $f_c = \lambda/2\pi \propto (N - N_c)^2$ *reddens* toward ungrounding (the spectral early-warning of Kuehn 2011; Bury et al. 2020). Crucially the fluctuation–dissipation identity $\mathrm{Var} \cdot 2\pi f_c = D$ holds independent of N, so the noise corner f_c is a **calibration-free** flotation-proximity gauge (it falls as $(N - N_c)^2$ regardless of the unknown noise strength D); combined with the tidal-admittance *response* (§4) it even returns D itself — a response/fluctuation pair from one velocity record. (Derived consistency relations, verified on OU realizations: corner recovered to 3%, FDT product constant to 5×10^{-17}; `nr3_misi_fold_ews`, `nr8_spectral_fdt_ews`.)

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6. The framework on the same real lakes

Reading open CryoSat-2 drainage + ITS_LIVE velocity (3 MacAyeal lakes) through this lens (`validation/external/lake_lag_sn_ews.py`, 5 tests): across the 3 resolved drainage events $|\Delta v/v| \leq$ 2%, mixed sign, near the ~1 % year-to-year noise floor — **0 positive >2σ surge detections** — so via the master curve these trunk lakes sit **far from the N_c pole**; and **no** lake shows the joint critical-slowing-down signature (rising variance *and* high AC1), so the early-warning correctly returns a **true negative** on stable ice (unit tests confirm the detectors *do* fire on planted surges/CSD, so the null is real). Both probes agree: far from ungrounding.

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7. Estimator calibration, scope, and falsification

7.1 Calibration. Every estimator has a synthetic plant-and-recover harness in `pytest`: the master-curve closed form (1.4×10^{-4}), the N_c inversion (0.2/0.4/1.0 %), the tidal admittance/harmonic recovery, and the CSD detectors (fire on planted surges/CSD, null on stable OU).

7.2 Scope and falsification. Absolute s_N is uncalibrated (per-event `dN/N` is a lumped-storage lower bound); the robust claims are the sign/shape and the *threshold* N_c. With n≈8 annual points per lake the CSD test is weak; a strong test needs the dense quarterly/GPS series and a stream actually nearing flotation. Falsifiers: a co-located `dN`+GPS population whose $|s_N(N)|$ does not follow $m/(1 - (N_c/N)^m)$; a stream nearing flotation with no variance/AC1 rise; tidal admittance/harmonics not steepening toward the grounding line.

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8. Reproduce

```
pip install -r requirements.txt
python glaciers/validation/synthetic/sn_master_curve.py       # §2 s_N(N) master curve + inversion
python glaciers/validation/synthetic/tidal_admittance_probe.py # §4 tidal third probe
python glaciers/validation/synthetic/spatial_ews.py           # §5 spatial early-warning
python glaciers/validation/external/lake_lag_sn_ews.py        # §6 framework on real lakes
pytest glaciers/tests/test_validation_synthetic.py -v         # §7 estimator calibration
```

Data and code availability

All analysis code and derived products are in the public repository at https://github.com/abdh0saifelden2-debug/K, and the results regenerate from the scripts above. The study uses the following open datasets:

- BedMachine Antarctica v4 (Morlighem, 2025, https://doi.org/10.5067/POJQI54A45HX; see Morlighem et al., 2020).
- CATS2008 circum-Antarctic tide model: USAP-DC 601235 (Howard et al., 2019, https://doi.org/10.15784/6012
- ITS_LIVE surface velocities (Gardner et al., 2025, https://doi.org/10.5067/JQ6337239C96; see Gardner et al., 2018) and CryoSat-2 lake-drainage dates.

BedMachine and ITS_LIVE were obtained from the NASA NSIDC DAAC; CATS2008 from the U.S. Antarctic Program Data Center (https://www.usap-dc.org).

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